

MD. TAJBIR HASAN SHUVO

AI/ML Engineer (Early Career)

Bogura, Bangladesh · +8801316334800 · tajshuvo75@gmail.com

[GitHub](#) | [LinkedIn](#) | Portfolio: tajshuvo.vercel.app

PROFESSIONAL SUMMARY

Early-career AI/ML Engineer specializing in LLM fine-tuning, model distillation, and multi-agent orchestration. Built reliability-aware RAG systems and deployed agentic AI pipelines with structured outputs and production ML APIs.

EDUCATION

Rajshahi University of Engineering & Technology (RUET), Bangladesh — BSc (Eng.) in Computer Science & Engineering

CGPA: 3.74 · March 2022 – August 2026

Thesis: "A Systematic Analysis of LoRA-Based Continual Learning for Vision Transformers in Class-Incremental Semantic Segmentation"

Bogura Cantonment Public School & College — HSC (GPA 5.00) · July 2018 – Nov 2020

RESEARCH EXPERIENCE

Research Intern, AIMS Lab — United International University (UIU) · July 2026 - Present

- Contributing to an LLM-based recruitment system automation platform, with a focus on fairness in candidate evaluation
- Designing a bias-mitigation engine to detect and reduce LLM bias across candidate gender, district, school, and language

TECHNICAL SKILLS

Machine Learning & Deep Learning: Supervised & Unsupervised Learning, Neural Networks (CNNs, LSTM), Transfer Learning, Model Distillation, Model Optimization (SGD, Adam), Regularization, Ensemble Methods

Transformers & LLM Systems: Transformer Architecture (Self-Attention, Multi-Head Attention, Encoder-Decoder), Language Modeling, Fine-Tuning Pretrained Models, LoRA / QLoRA / PEFT, RAG Pipelines, Hallucination Mitigation, Structured Output Enforcement, FAISS, Vector Search

Frameworks & Libraries: PyTorch, TensorFlow/Keras, Hugging Face Transformers, scikit-learn

Agentic AI & Orchestration: LangGraph (StateGraph, Multi-Stage Reasoning), LangChain, Multi-Agent Systems, Tool-Use Agents, MCP

Backend & Deployment: FastAPI, REST APIs, Streaming Inference, PostgreSQL, Docker

MLOps & Infrastructure: MLflow, DVC, Git, Linux/Bash, Kaggle Pipelines

Programming: Python (Primary), SQL, C/C++

PROJECTS

Brand Guardian AI | Python, FastAPI, LangGraph, Azure Video Indexer, Azure AI Search, RAG | [GitHub](#) | [Live](#)

- Architected an end-to-end agentic LLM pipeline for automated video advertisement compliance auditing from raw media input
- Orchestrated multi-stage reasoning using LangGraph over transcript + OCR extraction with semantic vector retrieval
- Implemented RAG-based policy evaluation generating evidence-grounded risk classification and structured compliance reports

Bangla LLM Distillation | Python, PyTorch, HuggingFace, LoRA/QLoRA, Transformers | [GitHub](#)

- Fine-tuned 7B Mistral-Instruct on 6K Bangla instructions (QLoRA, T4 GPU)
- Designed full LLM distillation pipeline compressing 7B → 1.1B (~6X smaller, ~25M trainable params)
- Achieved ~4.8x ROUGE-L gain over base 1.1B and retained ~50% teacher performance

Blog Writing Multi-Agent System | Python, LangGraph, LangChain, PostgreSQL (NeonDB), Streamlit | [GitHub](#) | [Live](#)

- Built a production-grade multi-agent LLM system with dynamic routing (closed/hybrid/open-book) and parallel section synthesis
- Enforced deterministic structured outputs via Pydantic schemas and retry-safe parsing to eliminate malformed LLM responses
- Integrated tool-based web research, citation control, and PostgreSQL JSONB persistence, deploying a Streamlit interface

Constitutional Reliability RAG (Constitution of Bangladesh) | Python, FastAPI, LangGraph, FAISS, RAG | [GitHub](#) | [Live](#)

- Built a reliability-aware agentic RAG system with FAISS retrieval, citation enforcement, faithfulness scoring, and self-correction
- Designed a LangGraph multi-stage reasoning pipeline with Pydantic state and source-grounded constitutional article validation
- Deployed a production-grade legal QA API (FastAPI) with streaming inference and reliability metrics for hallucination control

LANGUAGES

Bangla (Native) · English (Fluent)